Scenario-Based Conformal Prediction with Exact Conditional Coverage Guarantees

Guillaume O. Berger[1], Raphaël M. Jungers[1], and Licio Romao[2]

***Abstract*— Conditional coverage guarantees control the probability that a prediction set fails to attain a prescribed coverage level. We develop a family of conformal predictors based on convex scenario optimization that combines constraint removal with residual thresholding, without data splitting. Under convexity and atomless fixed-parameter residual distributions, we prove an exact Beta law for their coverage. The predictors are intersections of sets obtained from a chosen number of optimization stages, ranging from a single stage with residual thresholding to a cascade of stages with constraint removal. This construction allows a tradeoff between computation time and prediction-set volume while preserving the same coverage distribution for a fixed total batch size. Numerical experiments demonstrate substantial computational savings and smaller prediction-set volumes relative to the final-set removal baseline, depending on the chosen tradeoff.**

I. Introduction

Conformal prediction (CP) builds prediction sets for random data points based on finitely many observed data points [1], [2], [3]. The coverage of the prediction set is the probability that a new data point belongs to it. Under the assumption of exchangeability (which includes i.i.d. distribution of the data), conformal predictors come with so-called marginal coverage guarantees, that is, bounds on the expected value of the coverage of the prediction set obtained from data. Such guarantees are useful when the prediction set is computed several times with different data sets, because the average coverage then converges to its expected value. Nevertheless, in several applications, including safe control [4], machine learning [5], and operations research [6], the prediction set is computed once and used several times. In this case, one needs so-called conditional coverage guarantees, that is, bounds on the probability that the coverage exceeds some given threshold $1-\varepsilon$.

An early approach to CP with conditional coverage guarantees is split CP [7], where each calibration point is scored independently using a fixed nonconformity function (possibly fitted on a separate training data set). A key limitation of split CP is the tradeoff imposed by data splitting: fewer calibration observations can increase coverage variability, while fewer training observations can impair the quality of the fitted score function and lead to larger prediction sets [8].

Scenario optimization [9], [10], [11], [12], [13], [14] provides another route to CP with conditional coverage guarantees [8], [15]. The RemMCP method of [8] avoids data splitting by using the cascade removal scheme of [14], obtaining tight coverage guarantees at the cost of repeated optimization solves. Its constraint-relaxation counterpart reduces this computational burden but gives looser guarantees. In [15], scenario optimization selects a threshold for a fixed scalar score function after discarding a prescribed number of the largest sampled scores. Under i.i.d. sampling and a continuous score distribution, the resulting prediction set has an exact Beta coverage law. Risk allocation provides coverage guarantees for intersections of such sets, but can introduce conservatism. An earlier connection between CP and scenario optimization, focused on marginal coverage, appears in [16].

Contributions

We introduce a family of scenario-based conformal predictors that combines residual thresholding with cascade removal. Whereas RemMCP [8] removes constraints and refits the optimizer, our approach can tighten the prediction set while keeping the optimizer fixed. Specifically, we enlarge the batch of support observations with those having the largest remaining constraint residuals and use the smallest residual in this batch as the prediction threshold, instead of zero. This residual-based padding requires no additional optimization solve and can be used alone or at successive stages of a removal cascade. The construction of the predictor requires only L optimization solves, where L is the chosen number of stages.

Under convexity of the optimization program and atomless fixed-parameter residual distribution, we prove that the coverage of the prediction set follows exactly a $\mathrm{Beta}(N-m+1,m)$ distribution, where m is the total batch size and every stage has batch size at least d, the optimization dimension. Different allocations between padding and removal provide the same conditional coverage guarantee. The proof constructs a proper projective boundary scheme (see [17]) for the prediction set.

Choosing batches of size d recovers the intersection of the intermediate prediction sets in the support-removal cascade [14]. This intersection is contained in the final set returned by RemMCP [8] and can therefore reduce its volume while preserving the same coverage bound. The use of information from intermediate stages already appears in the compression scheme of [14, Section V-A]; our contribution is to incorporate residual padding and these intersections into a conformal predictor with an exact coverage law. Our numerical experiments illustrate the resulting tradeoff: the

[1]Guillaume O. Berger and Raphaël M. Jungers are with the Department of Applied Mathematics and the ICTEAM institute, UCLouvain, Louvain-la-Neuve, Belgium. Emails: guillaume.berger@uclouvain.be; raphael.jungers@uclouvain.be.

[2]Licio Romao is with the Department of Wind and Energy Systems, Technical University of Denmark, Lyngby, Denmark. Email: licio@dtu.dk.

residual thresholding approach substantially reduces fitting time compared to RemMCP, while retaining comparable prediction-set areas. The full cascade intersection consistently yields smaller areas than both RemMCP and pure residual thresholding. Furthermore, the proposed predictors closely track the Beta reference, while RemMCP exhibits more conservative coverage.

In comparison with the modular intersections of [15], our construction permits vector-valued optimization decisions and certifies the cascade intersection jointly, yielding an exact coverage law without allocating separate risk and confidence budgets to its components.

Unlike split CP [7], our method uses the full sample both to fit the scenario optimizer and to determine the residual threshold. The fitted scores are consequently dependent, and multiple active constraints produce ties at zero even for continuously distributed observations. Our exact coverage result therefore does not follow from the continuous-score equality in [7, Proposition 2b]. The atomlessness assumption concerns residuals at each fixed decision, rather than continuity of the jointly fitted scores.

Residual thresholding is also connected to [18], which derives exact risk distributions for cost thresholds chosen from the sampled costs after solving a scenario min–max problem. The optimizer remains fixed as the threshold varies; removing constraints and solving the problem again would generally change the optimizer. We extend this thresholding principle to convex scenario programs with a general convex objective and constraint function, and combine it with cascade removal and conformal ranking.

All proofs are in the appendices.

Notation. For a set S, S^* is the set of finite tuples with elements in S, including the empty tuple. For two sets A and B, $A \triangle B$ denotes their symmetric difference. $\mathbb{N}$ is the set of nonnegative integers, and $\mathbb{N}_* = \mathbb{N} \setminus \{0\}$.

II. Background and Problem Statement

A. Conformal Prediction and Coverage Guarantees

Let $\mathcal{Z}$ be a measurable space endowed with an unknown probability distribution $\mathbb{P}$. Define a *set predictor* $\Gamma : \mathcal{Z}^* \to 2^{\mathcal{Z}}$, mapping finite sequences on $\mathcal{Z}$ to measurable subsets of $\mathcal{Z}$. For a sequence $\boldsymbol{z} \in \mathcal{Z}^*$, define the coverage of its associated prediction set by

$$C(\boldsymbol{z}) = \mathbb{P}[\Gamma(\boldsymbol{z})].$$

We assume that C is measurable on $\mathcal{Z}^n$ for every sample size n. Thus, the coverage is the probability that an independent future observation drawn from $\mathbb{P}$ belongs to $\Gamma(\boldsymbol{z})$. Before observing the sample $\boldsymbol{z}$, coverage is a random variable.

In this paper, we always assume that $\boldsymbol{z}$ is sampled i.i.d. from $\mathbb{P}$. For every sample size n, we denote by $\mathbb{P}^n$ the associated distribution on $\mathcal{Z}^n$, i.e., that of an i.i.d. sample of size n.

Our focus is on *conditional coverage guarantees*, which quantify the probability that a prediction set computed from a sample of size N has sufficiently high coverage. Specifically, we study guarantees of the form[1]

$$\mathbb{P}^N\{C(\boldsymbol{z}) < 1 - \varepsilon\} \le \delta, \tag{1}$$

where $1 - \varepsilon \in (0, 1)$ is the required coverage level and $\delta \in [0, 1]$ is the failure probability over sampled data sets. Equivalently, the prediction set has coverage at least $1 - \varepsilon$ with probability at least $1 - \delta$. The bound is uniform over all distributions satisfying the stated assumptions; δ may depend on N and ε, but not on $\mathbb{P}$.

For later use, for integers $1 \le s \le n$ and $\varepsilon \in [0, 1]$, define the binomial lower tail

$$B_{n,s}(\varepsilon) = \sum_{j=0}^{s-1} \binom{n}{j} \varepsilon^j (1 - \varepsilon)^{n-j}. \tag{2}$$

Equivalently, if Z has distribution $\mathrm{Beta}(n - s + 1, s)$, then $\mathbb{P}\{Z < 1 - \varepsilon\} = B_{n,s}(\varepsilon)$ [19, Equation (8.17.5)].

Conformal prediction constructs prediction sets by ranking *nonconformity scores*, which measure how atypical an observation is relative to the data; larger scores indicate poorer conformity [2], [3]. For example, in regression, a score may be the absolute prediction error.

Definition 1. For each $n \in \mathbb{N}_*$, a *nonconformity scoring rule* is a measurable map $\mathcal{A}_n : \mathcal{Z}^n \to \mathbb{R}^n$ satisfying permutation equivariance: for every $\boldsymbol{u} = (u_1, \ldots, u_n)$ and permutation π of $\{1, \ldots, n\}$,

$$[\mathcal{A}_n(u_{\pi(1)}, \ldots, u_{\pi(n)})]_i = [\mathcal{A}_n(\boldsymbol{u})]_{\pi(i)}, \qquad 1 \le i \le n.$$

Given a sample $\boldsymbol{z} = (z_1, \ldots, z_N)$, the augmented scores associated with a candidate $z \in \mathcal{Z}$ are defined by

$$S_i(z) = [\mathcal{A}_{N+1}(z_1, \ldots, z_N, z)]_i, \qquad 1 \le i \le N + 1, \tag{3}$$

with $S_{N+1}(z)$ the candidate's score.[2] The conformal p-value of z (relative to $\boldsymbol{z}$) is defined by

$$p_{\boldsymbol{z}}(z) = \frac{1}{N+1} \sum_{i=1}^{N+1} \mathbf{1}\{S_i(z) \ge S_{N+1}(z)\}. \tag{4}$$

For a prescribed miscoverage level $\alpha \in (0, 1)$, the associated set predictor is

$$\Gamma_\alpha(\boldsymbol{z}) = \{z \in \mathcal{Z} : p_{\boldsymbol{z}}(z) > \alpha\}. \tag{5}$$

Whenever the sample $\boldsymbol{z}$ is i.i.d., the conformal predictor Γ_α satisfies the *marginal coverage guarantee* $\mathbb{E}[C(\boldsymbol{z})] \ge 1 - \alpha$, where the expectation is over the sample $\boldsymbol{z} \sim \mathbb{P}^N$; see [1]. This justifies the name "miscoverage level" for α since it controls the average miscoverage over sampled data sets. A conditional guarantee such as (1) requires additional analysis of the set predictor, which is the focus of this paper.

[1] Here, conditioning refers to the observed sample; in supervised prediction, it does not mean conditioning on each covariate value.

[2] Split CP is a subclass of CP where nonconformity scores are defined from an element-wise score function $h : \mathcal{Z} \to \mathbb{R}$: $[\mathcal{A}_n(\boldsymbol{u})]_i = h(u_i)$.

B. Convex Scenario Optimization

Scenario optimization, introduced in [9], [10], [11], samples realizations of a given probabilistic constraint and solves the associated optimization problem.

Concretely, on a domain $Q \subseteq \mathbb{R}^d$, consider the constraint $g(q; z) \leq 0$ on the variable q. This constraint is probabilistic in that it depends on z, which is our random observation.

For a sample $\boldsymbol{z} = (z_1, \ldots, z_n)$ and a set $Q \subseteq \mathbb{R}^d$, consider the associated *scenario program*

$$\begin{aligned} q_{\boldsymbol{z}} = \; & \underset{q \in Q}{\arg\min}\, J(q) \\ & \text{subject to } g(q; z_i) \leq 0, \quad 1 \leq i \leq n, \end{aligned} \tag{6}$$

where $J : Q \to \mathbb{R}$ is the *objective* and $g : Q \times \mathcal{Z} \to \mathbb{R}$ is the *constraint function*. We make the following standing assumption:

Assumption 1. The domain Q is convex and non-empty. The objective J is convex, and the constraint function $g(\cdot; z)$ is convex for every $z \in \mathcal{Z}$. For every $\boldsymbol{z} \in \mathcal{Z}^*$, the optimal solution $q_{\boldsymbol{z}}$ exists and is unique. The maps $\boldsymbol{z} \mapsto q_{\boldsymbol{z}}$ and $(q, z) \mapsto g(q, z)$ are measurable.

III. Proposed Scenario-Based Conformal Predictor

In this section, we propose a conformal set predictor based on nonconformity scores constructed from (6).

For a sample $\boldsymbol{z} = (z_1, \ldots, z_n)$, define the scoring rule

$$[\mathcal{A}_n(\boldsymbol{z})]_i = g(q_{\boldsymbol{z}}; z_i), \qquad 1 \leq i \leq n,$$

where $q_{\boldsymbol{z}}$ is defined in (6). In other words, an observation's score measures its constraint value at the optimizer computed with the full sample $\boldsymbol{z}$. Uniqueness of the optimizer makes $\mathcal{A}_n$ permutation equivariant. Feasibility makes all augmented scores nonpositive.

Given a sample $\boldsymbol{z} = (z_1, \ldots, z_N)$, the augmented scores associated with a candidate $z \in \mathcal{Z}$ are defined as in (3):

$$\begin{aligned} S_i(z) &= g(q_{\boldsymbol{z}z}; z_i), \qquad 1 \leq i \leq N, \\ S_{N+1}(z) &= g(q_{\boldsymbol{z}z}; z). \end{aligned} \tag{7}$$

For a prescribed miscoverage level $\alpha \in (0, 1)$, the associated set predictor defined in (5) with scores as in (7) is denoted by $\check{\Gamma}_\alpha$. Note that the nominal level α affects only the score ranking, not the optimization problem.

We make the following regularity assumption on the distribution $\mathbb{P}$, which will be used in subsequent developments (in particular, Theorem 1 and Theorem 2).

Assumption 2 (Atomless fixed-parameter score distribution)**.** For every fixed $q \in Q$ and $t \in \mathbb{R}$,

$$\mathbb{P}\{z \in \mathcal{Z} : g(q; z) = t\} = 0.$$

Assumption 2 concerns the residual distribution at each fixed parameter; it is stronger than atomlessness of the observations and does not assert independence of fitted scores.

A key lemma on score regularity under Assumption 2 is presented in Appendix II. Building on this lemma, the next subsections establish two key properties of $\check{\Gamma}_\alpha$: computational tractability and conditional coverage guarantees.

A. Computational Tractability: A Single Optimization

We show that given a sample $\boldsymbol{z} = (z_1, \ldots, z_N)$ and a candidate $z \in \mathcal{Z}$, the membership query $z \in \check{\Gamma}_\alpha(\boldsymbol{z})$ can be decided by solving (6) once, and then checking a function evaluation at z. Importantly, (6) need not be solved multiple times if other candidates are tested for membership.

To see this, let $q_{\boldsymbol{z}}$ be the solution of (6) (this is the only required solve of (6)) and write $S_i = g(q_{\boldsymbol{z}}; z_i)$. Order these scores as $S_{[1]} \geq \cdots \geq S_{[N]}$, and for $1 \leq r \leq N$ define

$$A_r(\boldsymbol{z}) = \{z \in \mathcal{Z} : g(q_{\boldsymbol{z}}; z) < S_{[r]}\}. \tag{8}$$

Note that once $q_{\boldsymbol{z}}$ and $S_{[r]}$ are computed, checking if $z \in A_r(\boldsymbol{z})$ reduces to computing $g(q_{\boldsymbol{z}}; z)$, which is in general much simpler than solving (6).

Theorem 1 (Reduction to one optimization)**.** *Consider program* (6) *under Assumptions 1–2. Let $N \in \mathbb{N}$, $\alpha \in (0, 1)$, and $r = \lfloor \alpha(N+1) \rfloor$. Assume that $d \leq r \leq N$. Almost surely on $\boldsymbol{z} \sim \mathbb{P}^N$, the predictors agree up to a $\mathbb{P}$-null set:*

$$\mathbb{P}\big[\check{\Gamma}_\alpha(\boldsymbol{z}) \triangle A_r(\boldsymbol{z})\big] = 0, \qquad \text{a.s. on } \boldsymbol{z} \sim \mathbb{P}^N. \tag{9}$$

B. Exact Conditional Coverage: A Beta Law

We now identify the coverage distribution of $\check{\Gamma}_\alpha$, proving that it follows a Beta law. The key is that A_r has the same coverage and admits a proper projective boundary of size r, allowing us to apply Theorem 7 (Appendix VII).

Theorem 2 (Beta distribution of coverage)**.** *Consider program* (6) *under Assumptions 1–2. Let $N \in \mathbb{N}$, $\alpha \in (0, 1)$, and $r = \lfloor \alpha(N+1) \rfloor$. Assume that $d \leq r \leq N$. For $\boldsymbol{z} \sim \mathbb{P}^N$, the coverage $\check{C}_\alpha(\boldsymbol{z}) = \mathbb{P}[\check{\Gamma}_\alpha(\boldsymbol{z})]$ satisfies*

$$\check{C}_\alpha(\boldsymbol{z}) \sim \mathrm{Beta}(N - r + 1, r). \tag{10}$$

Consequently, for every $\varepsilon \in (0, 1)$,

$$\mathbb{P}^N\{\check{C}_\alpha(\boldsymbol{z}) < 1 - \varepsilon\} = B_{N,r}(\varepsilon).$$

Remark 1 (The case of $r < d$)**.** Theorem 2 requires $r \geq d$. When $r = 0$, it holds that $\alpha < 1/(N + 1)$. Hence, every conformal p-value $p_{\boldsymbol{z}}(z)$ exceeds α, so that $\check{\Gamma}_\alpha(\boldsymbol{z}) = \mathcal{Z}$. For $1 \leq r < d$ and $N \geq d$, nesting of the conformal sets[3] and Theorem 2 at rank d give, for $\varepsilon \in [0, 1]$,

$$\mathbb{P}^N\{\check{C}_\alpha(\boldsymbol{z}) < 1 - \varepsilon\} \leq B_{N,d}(\varepsilon). \tag{11}$$

If every augmented problem has exactly d active constraints almost surely, every candidate's p-value is at least $d/(N+1)$ almost surely under $\mathbb{P}^{N+1}$. In that case, $\check{C}_\alpha(\boldsymbol{z}) = 1$ almost surely whenever $r < d$.

[3] I.e., $\alpha_1 < \alpha_2$ implies $\Gamma_{\alpha_1}(\boldsymbol{z}) \supseteq \Gamma_{\alpha_2}(\boldsymbol{z})$.

Algorithm 1: Cascade removal

Input: Sample $\boldsymbol{z} = (z_1, \dots, z_N)$ and positive integers $\boldsymbol{s} = (s_k)_{k=1}^{\infty}$.

$I_1 \leftarrow \{1, \dots, N\}$;
for $k = 1, 2, \dots$ **do**
  Solve (6) on $\boldsymbol{z}_{I_k}$ to obtain $q_{\boldsymbol{z}_{I_k}}$;
  if $|I_k| \geq s_k$ **then**
    $t_k \leftarrow s_k$th largest score $g(q_{\boldsymbol{z}_{I_k}}; z_i)$, $i \in I_k$;
  else
    $t_k \leftarrow -\infty$;
  $R_k \leftarrow \{i \in I_k : g(q_{\boldsymbol{z}_{I_k}}; z_i) \geq t_k\}$;
  $I_{k+1} \leftarrow I_k \setminus R_k$;

C. Extension to Cascade Removal

We extend the scenario-based conformal predictor introduced above to include removal of observations, following a cascade removal scheme inspired by [14], which is recalled in Appendix VIII.

Fix a sequence of positive integers $\boldsymbol{s} = (s_k)_{k=1}^{\infty}$. On input $\boldsymbol{z} = (z_1, \dots, z_N)$, consider the iterative removal process in Algorithm 1. An illustration of the algorithm with $s_k = 4$ is given in Appendix XI.

Given a run of Algorithm 1, we call R_k a *complete batch* when $|I_k| \geq s_k$, so that its threshold t_k is an order statistic and $|R_k| \geq s_k$; ties may increase its size. Note that if $|I_\ell| < s_\ell$, then trivially $I_k = \varnothing$ and $t_k = -\infty$ for all $k > \ell$. Hence, the algorithm runs only a finite number of nontrivial steps.

The above process allows us to define a nonconformity scoring rule: for every $\boldsymbol{z} = (z_1, \dots, z_n)$, define

$$[\mathcal{A}_n(\boldsymbol{z})]_i = g\big(q_{\boldsymbol{z}_{I_k}}; z_i\big) - Mk, \tag{12}$$

where k is the loop index such that $i \in R_k$ in the process, and $M > 0$ is a sufficiently large constant such that $g(q, z) > -M$ for all $q \in Q$ and $z \in \mathcal{Z}$.[4] Hence, earlier removals receive larger scores, while observations removed during the same round are ranked based on the constraint residual.

Given a sample $\boldsymbol{z} = (z_1, \dots, z_N)$, the augmented scores associated with a candidate $z \in \mathcal{Z}$ are defined as in (3):

$$S_i(z) = [\mathcal{A}_{N+1}(\boldsymbol{z}z)]_i, \tag{13}$$

with $\mathcal{A}_n$ as in (12). For a prescribed miscoverage level $\alpha \in (0, 1)$, the associated set predictor defined in (5) with scores as in (13) is denoted by $\bar{\Gamma}_{\boldsymbol{s},\alpha}$. Note that the nominal level α affects only the score ranking, not the optimization problem or the removal process.

The following result extends Theorem 1 to cascade removal, reducing the construction of the predictor to L optimization problems solved on the observed sample.

Theorem 3 (Reduction to L optimizations)**.** *Consider program* (6) *under Assumptions 1–2. Fix $N \in \mathbb{N}_*$, $\alpha \in (0, 1)$, and a sequence of positive integers $\boldsymbol{s} = (s_k)_{k=1}^{\infty}$ with $s_k \geq d$ for every k. Suppose that, for some $L \geq 1$,*

$$m \doteq \sum_{k=1}^{L} s_k = \lfloor \alpha(N+1) \rfloor.$$

Given $\boldsymbol{z} \in \mathcal{Z}^N$, run Algorithm 1 on $\boldsymbol{z}$ to obtain the index sets I_k and thresholds t_k for $1 \leq k \leq L$. Define[5]

$$G_{\boldsymbol{s},L}(\boldsymbol{z}) = \bigcap_{k=1}^{L} A_{s_k}(\boldsymbol{z}_{I_k}), \quad \textit{where}$$
$$A_{s_k}(\boldsymbol{z}_{I_k}) = \big\{z \in \mathcal{Z} : g\big(q_{\boldsymbol{z}_{I_k}}; z\big) < t_k\big\}. \tag{14}$$

Almost surely under $\mathbb{P}^N$, it holds that

$$\mathbb{P}\big[\bar{\Gamma}_{\boldsymbol{s},\alpha}(\boldsymbol{z}) \triangle G_{\boldsymbol{s},L}(\boldsymbol{z})\big] = 0.$$

Below we show that the exact conditional coverage guarantee extends to the conformal predictor with removal:

Theorem 4 (Beta distribution of coverage with removal)**.** *Consider program* (6) *under Assumptions 1–2. Fix $N \in \mathbb{N}_*$, $\alpha \in (0, 1)$, and a sequence of positive integers $\boldsymbol{s} = (s_k)_{k=1}^{\infty}$ with $s_k \geq d$ for every k. Suppose that, for some $L \geq 1$,*

$$m \doteq \sum_{k=1}^{L} s_k = \lfloor \alpha(N+1) \rfloor.$$

For $\boldsymbol{z} \sim \mathbb{P}^N$, the coverage $\bar{C}_{\boldsymbol{s},\alpha}(\boldsymbol{z}) = \mathbb{P}[\bar{\Gamma}_{\boldsymbol{s},\alpha}(\boldsymbol{z})]$ satisfies

$$\bar{C}_{\boldsymbol{s},\alpha}(\boldsymbol{z}) \sim \mathrm{Beta}(N - m + 1, m). \tag{15}$$

Consequently, for every $\varepsilon \in (0, 1)$,

$$\mathbb{P}^N\{\bar{C}_{\boldsymbol{s},\alpha}(\boldsymbol{z}) < 1 - \varepsilon\} = B_{N,m}(\varepsilon).$$

The proposed predictors thus share an exact coverage law determined by the total batch size m. The next section compares their computational cost and prediction-set size with those of the RemMCP baseline.

IV. Numerical Experiments

We compare the three predictor constructions on a simulated vehicle benchmark adapted from the car-trajectory example considered in [8]. The comparison examines coverage, prediction-set area, and fitting time at a common theoretical coverage reference. Data-generation details are available in Appendix IX, and all figures discussed below are in Appendix X.

A. Experimental Setup

The benchmark produces 2D data points whose distribution can be tuned with a parameter T (corresponding to the simulation horizon for the car trajectory). To compare the CP methods on data with different distributions, we therefore use three values of T, namely $T \in \{2, 30, 50\}$.

In this experiment, the scenario program (6) computes the smallest enclosing disk of the input sample $\boldsymbol{z} = (z_1, \dots, z_n)$.

[4] If such a constant does not exist, one can use a lexicographic order instead.

[5] Note that $A_{s_k}(\boldsymbol{z}_{I_k}) = \varnothing$ if $|I_k| < s_k$; otherwise, it coincides with (8).

TABLE I

MEANS OVER 100 RUNS. FOR THE INTERSECTION PREDICTORS, THE SUBSCRIPT DENOTES THE NUMBER OF STAGES: M3 USES $(3,3,3,3,3,3)$ AND M4 USES $(9,9)$. *For $T=2$, the area values are scaled by 10^5.*

T	Method	Coverage	Area	Computation (ms)
2	M1 ($\hat{\Gamma}_\ell$)	0.92535	3.225	66.08
	M2 ($\check{\Gamma}_\alpha$)	0.91069	4.860	2.28
	M3 ($\bar{\Gamma}_{\boldsymbol{s},\alpha}$)	0.91118	2.887	66.17
	M4 ($\bar{\Gamma}_{\boldsymbol{s},\alpha}$)	0.91072	3.575	19.53
30	M1 ($\hat{\Gamma}_\ell$)	0.92987	5.883	58.67
	M2 ($\check{\Gamma}_\alpha$)	0.91283	5.940	2.51
	M3 ($\bar{\Gamma}_{\boldsymbol{s},\alpha}$)	0.91107	5.351	60.93
	M4 ($\bar{\Gamma}_{\boldsymbol{s},\alpha}$)	0.91291	5.607	19.63
50	M1 ($\hat{\Gamma}_\ell$)	0.92638	56.695	60.93
	M2 ($\check{\Gamma}_\alpha$)	0.91293	56.193	2.46
	M3 ($\bar{\Gamma}_{\boldsymbol{s},\alpha}$)	0.90994	53.157	63.68
	M4 ($\bar{\Gamma}_{\boldsymbol{s},\alpha}$)	0.91191	54.753	21.19

Its decision dimension is $d = 3$.[6] For each $T \in \{2, 30, 50\}$ and for each of $n_{\mathrm{run}} = 100$ independent runs, we generate a sample $\boldsymbol{z}$ of $N = 200$ trajectories. All the compared methods (see below) use the same sample within a run.

We fix the target Beta distribution at $\mathrm{Beta}(N - r + 1, r)$ with $r = 18$ and compare four methods:

M1) *RemMCP (cascade removal)*, $\hat{\Gamma}_\ell$: the support-removal cascade approach of [14], used in [8], and recalled in Appendix VIII, with $\ell = 6$ rounds.
M2) *Residual thresholding*, $\check{\Gamma}_\alpha$: the residual thresholding approach in Section III, with $\alpha = 18/201$.
M3) *Six-stage generalized*, $\bar{\Gamma}_{\boldsymbol{s},\alpha}$: the generalized approach in Section III with $\boldsymbol{s} = (3,3,3,3,3,3)$ and $\alpha = 18/201$.
M4) *Two-stage generalized*, $\bar{\Gamma}_{\boldsymbol{s},\alpha}$: the generalized approach in Section III with $\boldsymbol{s} = (9,9)$ and $\alpha = 18/201$.

All methods have the reference distribution $\mathrm{Beta}(183, 18)$. For M2–M4, this is an exact coverage distribution under Assumption 2. For M1, the Beta CDF bounds the coverage CDF from above, under nondegeneracy.

For each run and method, we evaluated the following:

- *Coverage of the prediction set*: estimated by the fraction of $N_{\mathrm{test}} = 5000$ independent test observations that the prediction set contains.
- *Area of the prediction set*: computed by deterministic vertical-slice quadrature.
- *Computation time*: total computation time, including all scenario solves and predictor construction.

Table I summarizes the empirical coverage, prediction-set area, and computation time, averaged over n_{run} runs, for each method and horizon. Appendix X illustrates the fitted prediction sets and the distributions of coverage, prediction-set area, and computation time across runs.

[6]Two coordinates for the center and one variable controlling the radius.

B. Coverage and Prediction-Set Area

Across the three horizons, M2–M4 attain mean empirical coverages between 0.90994 and 0.91293, close to the mean $183/201 \approx 0.91045$ of the reference distribution $\mathrm{Beta}(183, 18)$. M1 attains higher mean empirical coverages, between 0.92535 and 0.92987, indicating conservatism relative to this reference, consistent with its one-sided coverage guarantee. Figure 2 extends the comparison to the distributions across runs: the empirical coverage CDFs of M2–M4 closely track the Beta reference, whereas that of M1 is shifted toward higher coverage.

Table I shows that M3 achieves the smallest mean area at all three horizons. Compared with M1, its mean area is smaller by 10.5%, 9.0%, and 6.2% at $T = 2, 30, 50$, respectively. Relative to M2, the corresponding reductions are 40.6%, 9.9%, and 5.4%; relative to M4, they are 19.3%, 4.6%, and 2.9%. Thus, among the proposed predictors M2–M4, using more stages yields smaller mean areas in this experiment at a common theoretical coverage distribution.

The comparison with M1 depends on the horizon. At $T = 2$, M1 has the second-smallest mean area, ahead of M4 and M2, despite its higher mean coverage. At $T = 30$, both M3 and M4 have smaller mean areas than M1, while M2 has a slightly larger mean area. At $T = 50$, all three proposed predictors have smaller mean areas than M1. These results show that higher coverage alone does not determine the ordering of prediction-set areas, since the fitted sets also differ in their geometry and location. Figure 3 complements these mean comparisons with the distributions across runs.

C. Computational Tradeoff

The computation times in Table I show the computational cost of the area reductions. M2, which solves a single scenario program, is fastest at every horizon, with mean computation times ranging from 2.28 to 2.51 ms. M4, which requires two scenario solutions, takes 19.53–21.19 ms, whereas M3 requires 60.93–66.17 ms, with six scenario solutions. M4 therefore provides an intermediate choice between the low computation cost of M2 and the smaller sets of M3, while preserving the same theoretical coverage distribution.

M1 and M3 both require six scenario solutions and have similar mean computation times: 58.67–66.08 ms for M1 versus 60.93–66.17 ms for M3. However, M3 yields smaller mean areas at every horizon and coverage closer to the reference mean. Figure 4 complements these mean comparisons with the variability of the computation times across runs.

CONCLUSION

We introduced a family of scenario-based conformal predictors combining residual thresholding and cascade removal without data splitting. Under the stated regularity assumptions, their coverage follows an exact Beta law determined by the sample size and total batch size, allowing the number of optimization stages to vary without changing the coverage guarantee. The vehicle-trajectory experiments illustrate the resulting tradeoff: a single optimization substantially reduces

computation time, while intersecting prediction sets across more stages yields smaller mean areas.

The proof uses Calafiore's projective-boundary framework [17]. We show that the selected batches form a proper projective boundary for the cascade intersection, whose size equals the total batch size almost surely. The framework then yields the exact Beta coverage law, certifying the intersection jointly without allocating risk separately to its components.

Future work could investigate how to choose the number of stages and allocate batch sizes to balance computation time and prediction-set volume. Choosing these quantities from the observed sample would require coverage guarantees that account for this additional dependence. Another direction is to extend the analysis to residual distributions with atoms, where ties may prevent an exact Beta law.

Use of AI. AI was used to verify proof correctness and completeness, leading to some minor fixes. We also used AI to help with the implementation of the code for the numerical experiments. All AI-generated outputs were thoroughly verified by the authors. The theoretical ideas originate entirely from the authors, without AI assistance.

REFERENCES

[1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic learning in a random world*. Springer, 2005.

[2] G. Shafer and V. Vovk, "A tutorial on conformal prediction," *Journal of Machine Learning Research*, vol. 9, pp. 371–421, 2008.

[3] A. N. Angelopoulos and S. Bates, "Conformal prediction: a gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.

[4] L. Lindemann, Y. Zhao, X. Yu, G. J. Pappas, and J. V. Deshmukh, "Formal verification and control with conformal prediction: practical safety guarantees for autonomous systems," *IEEE Control Systems*, vol. 45, no. 6, pp. 72–122, 2025.

[5] M. Bian and R. F. Barber, "Training-conditional coverage for distribution-free predictive inference," *Electronic Journal of Statistics*, vol. 17, no. 2, pp. 2044–2066, 2023.

[6] C. Johnstone and B. Cox, "Conformal uncertainty sets for robust optimization," in *Proceedings of the Tenth Symposium on Conformal and Probabilistic Prediction and Applications*, ser. Proceedings of Machine Learning Research, vol. 152, 2021, pp. 72–90.

[7] V. Vovk, "Conditional validity of inductive conformal predictors," in *Proceedings of the Asian Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 25, 2012, pp. 475–490.

[8] L. Lützow, S. Garatti, M. C. Campi, L. Lindemann, and M. Althoff, "Multi-variable conformal prediction: optimizing prediction sets without data splitting," arXiv:2605.12341, 2026. [Online]. Available: https://arxiv.org/abs/2605.12341

[9] G. Calafiore and M. C. Campi, "Uncertain convex programs: randomized solutions and confidence levels," *Mathematical Programming*, vol. 102, no. 1, pp. 25–46, 2005.

[10] M. C. Campi and S. Garatti, "The exact feasibility of randomized solutions of uncertain convex programs," *SIAM Journal on Optimization*, vol. 19, no. 3, pp. 1211–1230, 2008.

[11] G. C. Calafiore, "Random convex programs," *SIAM Journal on Optimization*, vol. 20, no. 6, pp. 3427–3464, 2010.

[12] M. C. Campi and S. Garatti, "A sampling-and-discarding approach to chance-constrained optimization: feasibility and optimality," *Journal of Optimization Theory and Applications*, vol. 148, no. 2, pp. 257–280, 2011.

[13] K. Margellos, M. Prandini, and J. Lygeros, "On the connection between compression learning and scenario based single-stage and cascading optimization problems," *IEEE Transactions on Automatic Control*, vol. 60, no. 10, pp. 2716–2721, 2015.

[14] L. Romao, A. Papachristodoulou, and K. Margellos, "On the exact feasibility of convex scenario programs with discarded constraints," *IEEE Transactions on Automatic Control*, vol. 68, no. 4, pp. 1986–2001, 2023.

[15] G. C. Calafiore, "Bridging conformal prediction and scenario optimization: discarded constraints and modular risk allocation," arXiv:2603.19396, 2026. [Online]. Available: https://arxiv.org/abs/2603.19396

[16] N. O'Sullivan, L. Romao, and K. Margellos, "Bridging conformal prediction and scenario optimization," in *2025 IEEE 64th Conference on Decision and Control (CDC)*, 2025, pp. 6114–6121.

[17] G. C. Calafiore, "Exact risk-complexity laws for projective boundaries in scenario optimization and distribution-free certification," arXiv:2609.01355, 2026. [Online]. Available: https://arxiv.org/abs/2609.01355

[18] A. Carè, S. Garatti, and M. C. Campi, "Scenario min-max optimization and the risk of empirical costs," *SIAM Journal on Optimization*, vol. 25, no. 4, pp. 2061–2080, 2015.

[19] National Institute of Standards and Technology, "NIST digital library of mathematical functions," section 8.17: Incomplete Beta Functions. Accessed October 11, 2026. [Online]. Available: https://dlmf.nist.gov/8.17

[20] R. Tumu, M. Cleaveland, R. Mangharam, G. Pappas, and L. Lindemann, "Multi-modal conformal prediction regions by optimizing convex shape templates," in *Proceedings of the 6th Annual Learning for Dynamics & Control Conference*, ser. Proceedings of Machine Learning Research, vol. 242. PMLR, 2024, pp. 1343–1356.

APPENDIX I
BACKGROUND ON CONVEX SCENARIO OPTIMIZATION

Definition 2 (Support index [10, Definition 2])**.** Let $\boldsymbol{z} \in \mathcal{Z}^n$. An index $1 \leq i \leq n$ is a *support index* for $\boldsymbol{z}$ (relative to (6)) if the removal of the ith element of $\boldsymbol{z}$ changes the optimizer, i.e., if $q_{\boldsymbol{z}_{\{1,\dots,n\}\setminus\{i\}}} \neq q_{\boldsymbol{z}}$.

We denote by $R(\boldsymbol{z})$ the set of support indices of $\boldsymbol{z}$.

Lemma 5 (Support bound [10, Proposition 1])**.** *Consider program* (6)*. For every $\boldsymbol{z} \in \mathcal{Z}^*$, it holds that $|R(\boldsymbol{z})| \leq d$.*

APPENDIX II
A KEY LEMMA ON SCORE REGULARITY

The lemma and its proof build on Appendix I.

Lemma 6 (Active constraints and ranks)**.** *Consider program* (6) *under Assumptions 1–2. Let $n \in \mathbb{N}$ and $\mathcal{N} = \{1, \dots, n\}$. The following holds almost surely on $\boldsymbol{z} \sim \mathbb{P}^n$:*

(i) *for all $i \in \mathcal{N}$, $g(q_{\boldsymbol{z}}; z_i) = 0$ if and only if $i \in R(\boldsymbol{z})$;*

(ii) *all scores $g(q_{\boldsymbol{z}}; z_i)$, $i \in \mathcal{N} \setminus R(\boldsymbol{z})$, are strictly negative and pairwise distinct.*

Consequently, for fixed $r \geq d$ and $n \geq r$, the indices of the r largest scores $g(q_{\boldsymbol{z}}; z_i)$, $i \in \mathcal{N}$, form a unique, permutation-equivariant set containing the support indices.

Proof. The proof of the "if" direction in (i) is intuitive, but a formal proof can be found in [10]. Next, for fixed $i \in \mathcal{N}$, the leave-one-out optimizer $q_{\boldsymbol{z}}^{\neg i}$ (solving (6) on $\boldsymbol{z}_{\mathcal{N}\setminus\{i\}}$) depends only on the observations $\boldsymbol{z}_{\mathcal{N}\setminus\{i\}}$ and is therefore independent of z_i. Writing ν_i for its law, Tonelli's theorem gives

$$\mathbb{P}^n\{g(q_{\boldsymbol{z}}^{\neg i}; z_i) = 0\} = \int_Q \mathbb{P}\{z : g(q; z) = 0\}\, \nu_i(dq) = 0.$$

On the event A_i that i is not a support index of $\boldsymbol{z}$, we have $q_{\boldsymbol{z}}^{\neg i} = q_{\boldsymbol{z}}$ by definition. Hence,

$$\mathbb{P}^n\{g(q_{\boldsymbol{z}}; z_i) = 0 \wedge A_i\} \leq \mathbb{P}^n\{g(q_{\boldsymbol{z}}^{\neg i}; z_i) = 0\} = 0.$$

A union bound over $i \in \mathcal{N}$ then proves the "only if" direction in (i) and strict negativity in (ii).

To prove pairwise distinctness in (ii), fix $i \neq j$ in $\mathcal{N}$, and denote by $q_{\boldsymbol{z}}^{\neg i,j}$ the leave-two-out optimizer (solving (6) on $\boldsymbol{z}_{\mathcal{N}\setminus\{i,j\}}$). On the event $B_{i,j}$ that $g(q_{\boldsymbol{z}}; z_i)$ and $g(q_{\boldsymbol{z}}; z_j)$ are strictly negative, deleting the ith observation in $\boldsymbol{z}$ leaves the optimizer unchanged by the argument above, i.e., $q_{\boldsymbol{z}}^{\neg i} = q_{\boldsymbol{z}}$. Hence, $g(q_{\boldsymbol{z}}^{\neg i}; z_j) < 0$, so deleting it also leaves the optimizer unchanged. Thus, $q_{\boldsymbol{z}}^{\neg i,j} = q_{\boldsymbol{z}}$ on $B_{i,j}$. For fixed $q \in Q$, let μ_q be the law of $g(q; z)$ for $z \sim \mathbb{P}$. Assumption 2 implies

$$(\mu_q \otimes \mu_q)\{(a,b) : a = b\} = \int_{\mathbb{R}} \mu_q(\{t\})\, \mu_q(dt) = 0.$$

Since $q_{\boldsymbol{z}}^{\neg i,j}$ is independent of z_i and z_j, integrating over $q_{\boldsymbol{z}}^{\neg i,j}$ shows that $\mathbb{P}^n\{g(q_{\boldsymbol{z}}^{\neg i,j}; z_i) = g(q_{\boldsymbol{z}}^{\neg i,j}; z_j)\} = 0$. Hence,

$$\mathbb{P}^n\{g(q_{\boldsymbol{z}}; z_i) = g(q_{\boldsymbol{z}}; z_j) \wedge B_{i,j}\} \leq \mathbb{P}^n\{g(q_{\boldsymbol{z}}^{\neg i,j}; z_i) = g(q_{\boldsymbol{z}}^{\neg i,j}; z_j)\} = 0.$$

Taking a union bound over $i \neq j$ in $\mathcal{N}$, we conclude that two strictly negative fitted scores cannot coincide except on a null event, proving (ii).

Finally, Lemma 5 gives that $|R(\boldsymbol{z})| \leq d \leq r$ for all $\boldsymbol{z} \in \mathcal{Z}^n$. All zero scores therefore belong to the top r, and the remaining scores are distinct with probability one, so no tie crosses the selection cutoff. The selected index set is unique and transforms equivariantly under permutations because the optimizer is permutation invariant, concluding the proof. □

Appendix III
Proof of Theorem 1

For every fixed $\boldsymbol{z}$, Assumption 2 gives $\mathbb{P}\{z : g(q_{\boldsymbol{z}}; z) = S_{[r]}\} = 0$. By Tonelli's theorem and Lemma 6, almost every augmented sample $\boldsymbol{z}z \sim \mathbb{P}^{N+1}$ therefore satisfies $g(q_{\boldsymbol{z}}; z) \neq S_{[r]}$ and the conclusions of that lemma. Fix such a sample.

If $g(q_{\boldsymbol{z}}; z) \leq 0$, uniqueness gives $q_{\boldsymbol{z}z} = q_{\boldsymbol{z}}$. Put $m(z) = |\{i \in \{1,\dots,N\} : S_i \geq g(q_{\boldsymbol{z}}; z)\}|$. Definition (4) then gives $p_{\boldsymbol{z}}(z) = (m(z)+1)/(N+1)$. Since $r = \lfloor \alpha(N+1) \rfloor$ and $m(z)$ is an integer,

$$\begin{aligned} z \in \check{\Gamma}_\alpha(\boldsymbol{z}) &\iff m(z) \geq r \\ &\iff g(q_{\boldsymbol{z}}; z) < S_{[r]} \\ &\iff z \in A_r(\boldsymbol{z}). \end{aligned}$$

The strict inequality follows from $g(q_{\boldsymbol{z}}; z) \neq S_{[r]}$.

If $g(q_{\boldsymbol{z}}; z) > 0$, the original optimizer is infeasible for the augmented problem. This implies that $q_{\boldsymbol{z}z} \neq q_{\boldsymbol{z}}$, so that $N+1$ is a support index of $\boldsymbol{z}z$. By hypothesis on $\boldsymbol{z}z$, all scores $S_i(z)$ are nonpositive, and the zero scores correspond exactly to support indices of $\boldsymbol{z}z$ (including $N+1$). Hence, by Lemma 5,

$$p_{\boldsymbol{z}}(z) = \frac{\sum_{i=1}^{N+1} \mathbf{1}\{S_i(z) \geq 0\}}{N+1} = \frac{|R(\boldsymbol{z}z)|}{N+1} \leq \frac{r}{N+1} \leq \alpha,$$

so $z \notin \check{\Gamma}_\alpha(\boldsymbol{z})$. Also $z \notin A_r(\boldsymbol{z})$, since $g(q_{\boldsymbol{z}}; z) > 0 \geq S_{[r]}$.

The above equivalence holds almost surely on $\boldsymbol{z}z \sim \mathbb{P}^{N+1}$. By Tonelli's theorem,

$$\int_{\mathcal{Z}^N} \mathbb{P}\big[\check{\Gamma}_\alpha(\boldsymbol{z}) \triangle A_r(\boldsymbol{z})\big]\, \mathbb{P}^N(d\boldsymbol{z}) = \mathbb{P}^{N+1}\big\{z \in \check{\Gamma}_\alpha(\boldsymbol{z}) \triangle A_r(\boldsymbol{z})\big\} = 0.$$

The integrand is nonnegative, proving (9).

Appendix IV
Proof of Theorem 2

By Theorem 1, $\check{C}_\alpha(\boldsymbol{z}) = \mathbb{P}[A_r(\boldsymbol{z})]$ almost surely on $\boldsymbol{z} \sim \mathbb{P}^N$. Following the approach of [17], summarized in Appendix VII, we will build a proper projective boundary scheme for A_r in order to apply Theorem 7 and prove the coverage law.

We keep r fixed throughout this proof, even when changing the sample size n. For a sample $\boldsymbol{z} = (z_1,\dots,z_n)$, with $n \geq r$, write $t_r(\boldsymbol{z})$ for the rth largest fitted score $g(q_{\boldsymbol{z}}; z_i)$, $1 \leq i \leq n$, and define the boundary

$$\mathcal{B}_r(\boldsymbol{z}) = \{i \in \mathcal{N} : g(q_{\boldsymbol{z}}; z_i) \geq t_r(\boldsymbol{z})\},$$

where $\mathcal{N} = \{1,\dots,n\}$. The boundary includes every tie at $t_r(\boldsymbol{z})$, so it is well defined and permutation equivariant. If $n < r$, set $A_r(\boldsymbol{z}) = \varnothing$ and $\mathcal{B}_r(\boldsymbol{z}) = \{1,\dots,n\}$. We verify the boundary identities (19) and (20) from Definition 3.

Let $\boldsymbol{z} = (z_1,\dots,z_n)$ and $I \subseteq \mathcal{N} \doteq \{1,\dots,n\}$. If $|I| < r$, (19) and (20) are trivially satisfied, from the above definitions. Hence, we focus on the case $|I| \geq r$.

1) Proof of (19): ($\Rightarrow$) Suppose that $\{z_j : j \in \mathcal{N} \setminus I\} \subseteq A_r(\boldsymbol{z}_I)$. Since $t_r(\boldsymbol{z}_I) \leq 0$, $q_{\boldsymbol{z}_I}$ satisfies $g(q_{\boldsymbol{z}_I}; z_i) \leq 0$ for all $i \in \mathcal{N} \setminus I$. Uniqueness of the optimizer implies $q_{\boldsymbol{z}} = q_{\boldsymbol{z}_I}$. Every $j \in \mathcal{N} \setminus I$ satisfies $g(q_{\boldsymbol{z}}; z_j) < t_r(\boldsymbol{z}_I)$, while at least r indices $i \in I$ satisfy $g(q_{\boldsymbol{z}}; z_i) \geq t_r(\boldsymbol{z}_I)$. Hence, $\mathcal{B}_r(\boldsymbol{z}) \subseteq I$.

($\Leftarrow$) Suppose $\mathcal{B}_r(\boldsymbol{z}) \subseteq I$. For every $j \in \mathcal{N} \setminus I$, it holds that $g(q_{\boldsymbol{z}}; z_j) < t_r(\boldsymbol{z}) \leq 0$. The same argument as in the proof of Lemma 6 (cf. the proof that $q_{\boldsymbol{z}}^{\neg i,j} = q_{\boldsymbol{z}}$) then shows that $q_{\boldsymbol{z}_I} = q_{\boldsymbol{z}}$. Since every $i \in \mathcal{N}$ with $g(q_{\boldsymbol{z}}; z_i) \geq t_r(\boldsymbol{z})$ belongs to $\mathcal{B}_r(\boldsymbol{z}) \subseteq I$, it holds that $t_r(\boldsymbol{z}_I) = t_r(\boldsymbol{z})$. Hence, for all $j \in \mathcal{N} \setminus I$, $z_j \in A_r(\boldsymbol{z}_I)$, proving (19).

2) Proof of (20): If $\mathcal{B}_r(\boldsymbol{z}) \subseteq I$, the preceding argument gives $q_{\boldsymbol{z}_I} = q_{\boldsymbol{z}}$ and $t_r(\boldsymbol{z}_I) = t_r(\boldsymbol{z})$. Consequently, $\mathcal{B}_r(\boldsymbol{z}_I) = \{\rho_I(i) : i \in \mathcal{B}_r(\boldsymbol{z})\}$, proving (20).

Thus, $(A_r, \mathcal{B}_r)$ is a proper projective boundary scheme as defined in Definition 3. Lemma 6 gives $|\mathcal{B}_r(\boldsymbol{z})| = r$ almost surely on $\boldsymbol{z} \sim \mathbb{P}^n$ for all $n \geq r$, since no tie occurs at $t_r(\boldsymbol{z})$. Theorem 7 gives the Beta law for $\mathbb{P}[A_r(\boldsymbol{z})]$. Theorem 1 transfers this law to $\check{C}_\alpha$.

Appendix V
Proof of Theorem 3

Put $b_0 = 0$ and $m = b_L = \lfloor \alpha(N+1) \rfloor$. Consider a random input $\boldsymbol{u} \sim \mathbb{P}^n$ to the process. Since $s_k \geq d$ for all k, Lemma 6, applied to all subcollections, gives that complete batches R_k have size s_k almost surely. Hence, since $N - b_{k-1} \geq s_k$ for $k \leq L$, it follows that if $n \geq N$, the first L batches are complete almost surely.

Apply the above almost-sure conclusions at sizes N and $N+1$. By Tonelli's theorem, this gives an almost sure set $E \subseteq \mathcal{Z}^N$ such that for every $\boldsymbol{z} \in E$ and almost every $z \sim \mathbb{P}$, the process satisfies these conclusions both on input $\boldsymbol{z}$ and on input $\boldsymbol{z}z$. Now, fix $\boldsymbol{z} \in E$. The optimizers $q_{\boldsymbol{z}_{I_k}}$ and the thresholds t_k are fixed. Assumption 2 therefore implies that almost surely on $z \sim \mathbb{P}$,

$$g\big(q_{\boldsymbol{z}_{I_k}}; z\big) \notin \{0, t_k\}, \qquad 1 \le k \le L.$$

Let $F \subseteq E \times \mathcal{Z}$ be the almost sure set of augmented inputs $\boldsymbol{z}z$ satisfying the preceding properties. Fix $\boldsymbol{z}z \in F$. Let R_k and $\tilde{R}_k$ denote the batches on inputs $\boldsymbol{z}$ and $\boldsymbol{z}z$, respectively. Similarly, let t_k denote the thresholds on input $\boldsymbol{z}$.

Suppose $\tilde{R}_j = R_j$ for $j < k$, so the remaining augmented sample is $\boldsymbol{z}_{I_k}z$. Write $v_k = g(q_{\boldsymbol{z}_{I_k}}; z)$. If $v_k < 0$, adding the candidate preserves the optimizer by uniqueness. Since $v_k \neq t_k$, comparison with the s_kth largest residual gives

$$v_k < t_k \implies \tilde{R}_k = R_k, \qquad v_k > t_k \implies N+1 \in \tilde{R}_k.$$

If $v_k > 0$, then adding the candidate changes the optimizer, making $N+1$ a support index of $\boldsymbol{z}_{I_k}z$. Lemma 6 gives it residual zero, shared by at most $d \le s_k$ observations; hence $N+1 \in \tilde{R}_k$ again. Induction therefore yields

$$z \in G_{\boldsymbol{s},L}(\boldsymbol{z}) \iff N+1 \notin \bigcup_{k=1}^{L} \tilde{R}_k.$$

By (12), every earlier batch has strictly larger scores than every later batch. Since $m = b_L$, the batches $\tilde{R}_k$, $1 \le k \le L$, contain exactly m indices. Thus, the candidate's total rank (counting all scores at least its own) is at most m if it belongs to one of these batches and at least $m+1$ otherwise. Since $m/(N+1) \le \alpha < (m+1)/(N+1)$, acceptance is equivalent to avoiding these batches, hence to membership in $G_{\boldsymbol{s},L}(\boldsymbol{z})$. This holds for every $\boldsymbol{z}z$ in the almost sure set $F \subseteq \mathcal{Z}^{N+1}$, proving the claim by Tonelli's theorem.

Appendix VI
Proof of Theorem 4

Keep $\boldsymbol{s}$ and L fixed throughout the proof, including when changing the sample size n. For every finite sample $\boldsymbol{z} = (z_1, \dots, z_n)$, use $G_{\boldsymbol{s},L}$ as in (14) and define the boundary

$$\mathcal{B}_{\boldsymbol{s},L}(\boldsymbol{z}) = \bigcup_{k=1}^{L} R_k. \tag{16}$$

Note that $A_{s_k} = \varnothing$ on samples shorter than s_k. By the proof of Theorem 2, each pair $(A_{s_k}, \mathcal{B}_{s_k})$, $1 \le k \le L$, is a proper projective boundary scheme, whose boundary $\mathcal{B}_{s_k}(\boldsymbol{z}_{I_k})$ is precisely R_k, up to reindexing. We show that $(G_{\boldsymbol{s},L}, \mathcal{B}_{\boldsymbol{s},L})$ is itself a proper projective boundary scheme.

Fix $\boldsymbol{z} = (z_1, \dots, z_n)$ and $I \subseteq \mathcal{N} \doteq \{1, \dots, n\}$. Denote by R_k and $\tilde{R}_k$ the batches on inputs $\boldsymbol{z}$ and $\boldsymbol{z}_I$, respectively, both expressed in $\boldsymbol{z}$'s indexing. Similarly, denote by I_k and $\tilde{I}_k$ the residual index sets. We verify the boundary identities (19) and (20).

1) Proof of (19): ($\Rightarrow$) Suppose that $z_j \in G_{\boldsymbol{s},L}(\boldsymbol{z}_I)$ for every $j \in \mathcal{N} \setminus I$. We prove by induction that $R_k = \tilde{R}_k \subseteq I$ for $1 \le k \le L$.

At $k = 1$, every such z_j belongs to $A_{s_1}(\boldsymbol{z}_I)$. Boundary equivalence for $(A_{s_1}, \mathcal{B}_{s_1})$ therefore gives $R_1 = \mathcal{B}_{s_1}(\boldsymbol{z}) \subseteq I$, and projectivity gives $R_1 = \mathcal{B}_{s_1}(\boldsymbol{z}) = \mathcal{B}_{s_1}(\boldsymbol{z}_I) = \tilde{R}_1$. Now, assume that $R_j = \tilde{R}_j \subseteq I$ for every $j < k$. Then

$$\tilde{I}_k = I \setminus \bigcup_{j<k} \tilde{R}_j = I_k \cap I, \qquad I_k \setminus I = \mathcal{N} \setminus I. \tag{17}$$

By assumption, $z_j \in A_{s_k}(\boldsymbol{z}_{\tilde{I}_k})$ for all $j \in \mathcal{N} \setminus I$. Hence, by the above, $z_j \in A_{s_k}(\boldsymbol{z}_{I_k \cap I})$ for all $j \in I_k \setminus I$. Boundary equivalence then gives $R_k = \mathcal{B}_{s_k}(\boldsymbol{z}_{I_k}) \subseteq I$. Projectivity then gives $R_k = \tilde{R}_k$, since after translating to $\boldsymbol{z}$'s indexing,

$$R_k = \mathcal{B}_{s_k}(\boldsymbol{z}_{I_k}) = \mathcal{B}_{s_k}(\boldsymbol{z}_{I_k \cap I}) = \mathcal{B}_{s_k}(\boldsymbol{z}_{\tilde{I}_k}) = \tilde{R}_k. \tag{18}$$

This completes the induction, so $\mathcal{B}_{\boldsymbol{s},L}(\boldsymbol{z}) = \bigcup_{k=1}^{L} R_k \subseteq I$.

($\Leftarrow$) Suppose that $\mathcal{B}_{\boldsymbol{s},L}(\boldsymbol{z}) = \bigcup_{k=1}^{L} R_k \subseteq I$. We prove by induction that $R_k = \tilde{R}_k \subseteq I$ for $1 \le k \le L$.

At $k = 1$, since $R_1 \subseteq I$ by hypothesis, projectivity gives $R_1 = \mathcal{B}_{s_1}(\boldsymbol{z}) = \mathcal{B}_{s_1}(\boldsymbol{z}_I) = \tilde{R}_1$. Now, assume that $R_j = \tilde{R}_j$ for every $j < k$, so that (17) holds. Since $R_k \subseteq I_k \cap I$ by hypothesis, projectivity gives $R_k = \tilde{R}_k$ for the same reason as in (18). This concludes the induction.

Now, fix $1 \le k \le L$. Since $R_k \subseteq I$, boundary equivalence gives $z_j \in A_{s_k}(\boldsymbol{z}_{I_k \cap I})$ for every $j \in I_k \setminus I$. Then, from (17), $z_j \in A_{s_k}(\boldsymbol{z}_{\tilde{I}_k})$ for every $j \in \mathcal{N} \setminus I$. Since k is arbitrary, it follows that $z_j \in G_{\boldsymbol{s},L}(\boldsymbol{z}_I)$ for every $j \in \mathcal{N} \setminus I$, proving (19).

2) Proof of (20): Suppose that $\mathcal{B}_{\boldsymbol{s},L}(\boldsymbol{z}) \subseteq I$. The induction in ($\Leftarrow$) gives $\tilde{R}_k = R_k$ for every $1 \le k \le L$, with both sets expressed in $\boldsymbol{z}$'s indexing. Translating their union to the indexing of $\boldsymbol{z}_I$ yields

$$\begin{aligned} \mathcal{B}_{\boldsymbol{s},L}(\boldsymbol{z}_I) &= \bigcup_{k=1}^{L} \{\rho_I(i) : i \in \tilde{R}_k\} \\ &= \{\rho_I(i) : i \in \mathcal{B}_{\boldsymbol{s},L}(\boldsymbol{z})\}, \end{aligned}$$

which proves projectivity.

For every $n \ge m$, Lemma 6, applied simultaneously to all subcollections of an i.i.d. sample of size n, implies that the first L batches have sizes $s_1, \dots, s_L$ almost surely. Since the batches are disjoint, $|\mathcal{B}_{\boldsymbol{s},L}(\boldsymbol{z})| = m$ almost surely for every $n \ge m$. Theorem 7 therefore gives

$$\mathbb{P}[G_{\boldsymbol{s},L}(\boldsymbol{z})] \sim \mathrm{Beta}(N - m + 1, m).$$

Theorem 3 transfers this law to $\bar{C}_{\boldsymbol{s},\alpha}$. The tail identity follows from (2).

Supplementary Material

Appendix VII
Proper projective boundaries

We present the notion of proper projective boundary from [17], which is instrumental in Appendices IV and VI.

Let $\Gamma : \mathcal{Z}^* \to 2^{\mathcal{Z}}$ be a permutation-invariant set predictor with a jointly measurable membership indicator.[7]

A *boundary map* is a measurable, permutation-equivariant map $\mathcal{B} : \mathcal{Z}^* \to 2^{\mathbb{N}}$ such that for all $\boldsymbol{z}$, $\mathcal{B}(\boldsymbol{z})$ is a subset of the indices of $\boldsymbol{z}$, i.e., if $\boldsymbol{z} \in \mathcal{Z}^n$, then $\mathcal{B}(\boldsymbol{z}) \subseteq \{1, \dots, n\}$. For $\boldsymbol{z} = (z_1, \dots, z_n)$ and $I \subseteq \{1, \dots, n\}$, let $\boldsymbol{z}_I = (z_i)_{i \in I}$, ordered by increasing index. For $i \in I$, define $\rho_I(i) = |\{j \in I : j \le i\}|$, the index of i within the ordered subcollection. The rules are also defined on the empty sample: $\mathcal{B}(\varnothing) = \varnothing$.

Definition 3 (Proper projective boundary scheme [17, Definition 3.4])**.** The pair $(\Gamma, \mathcal{B})$ is a *proper projective boundary scheme* if the following two conditions hold for every $\boldsymbol{z} = (z_1, \dots, z_n)$ and $I \subseteq \mathcal{N} \doteq \{1, \dots, n\}$:

$$\{z_j : j \in \mathcal{N} \setminus I\} \subseteq \Gamma(\boldsymbol{z}_I) \iff \mathcal{B}(\boldsymbol{z}) \subseteq I, \tag{19}$$

$$\mathcal{B}(\boldsymbol{z}) \subseteq I \implies \mathcal{B}(\boldsymbol{z}_I) = \{\rho_I(i) : i \in \mathcal{B}(\boldsymbol{z})\}. \tag{20}$$

Boundary equivalence (19) means that removing observations leaves all of them accepted by the remaining predictor if and only if the removed observations are not in the boundary. Projectivity (20) means that deleting observations outside this boundary preserves the boundary, up to reindexing. The following result will be used in the subsequent analysis.

Theorem 7 ([17, Corollaries 4.5 and 4.6])**.** *Let $(\Gamma, \mathcal{B})$ be a proper projective boundary scheme and fix $s \in \mathbb{N}_*$. Suppose that, for every $n \ge s$, $\mathbb{P}^n\{|\mathcal{B}(\boldsymbol{z})| \le s\} = 1$. Then, for every $N \ge s$ and $\varepsilon \in (0, 1)$,*

$$\mathbb{P}^N\{C(\boldsymbol{z}) < 1 - \varepsilon\} \le B_{N,s}(\varepsilon).$$

If additionally $\mathbb{P}^n\{|\mathcal{B}(\boldsymbol{z})| = s\} = 1$ for every $n \ge s$, then for every $N \ge s$, $C(\boldsymbol{z}) \sim \mathrm{Beta}(N - s + 1, s)$ when $\boldsymbol{z} \sim \mathbb{P}^N$, and the preceding bound holds with equality.

Appendix VIII
Convex scenario optimization with cascade removal

We recall the support-removal cascade of [14], used in [16], [8] to construct conformal predictors.

Definition 4 (Nondegeneracy [11, Definition 2.7])**.** Let $\boldsymbol{z} \in \mathcal{Z}^n$. We say that $\boldsymbol{z}$ is *nondegenerate* (relative to (6)) if its support elements reconstruct the optimizer, i.e., if $q_{\boldsymbol{z}_{R(\boldsymbol{z})}} = q_{\boldsymbol{z}}$.

Fix $\ell \in \mathbb{N}_*$. On input $\boldsymbol{z} = (z_1, \dots, z_N)$ with $N \ge \ell d$, the removal process is given in Algorithm 2.

[7] Permutation invariance means that reordering the sample leaves the prediction set unchanged: for every $\boldsymbol{u} = (u_1, \dots, u_n)$ and permutation π of $\{1, \dots, n\}$, $\Gamma(u_{\pi(1)}, \dots, u_{\pi(n)}) = \Gamma(\boldsymbol{u})$. Joint measurability means that $(\boldsymbol{z}, z) \mapsto \mathbf{1}\{z \in \Gamma(\boldsymbol{z})\}$ is measurable, ensuring that acceptance events have well-defined probabilities.

Algorithm 2: Support-removal cascade

Input: Sample $\boldsymbol{z} = (z_1, \dots, z_N)$ and number of rounds $\ell \in \mathbb{N}_*$, with $N \ge \ell d$.

$I_1 \leftarrow \{1, \dots, N\}$;
for $k = 1, \dots, \ell - 1$ **do**
  Solve (6) on $\boldsymbol{z}_{I_k}$ to obtain $q_{\boldsymbol{z}_{I_k}}$ and support indices $R_k \subseteq I_k$, expressed in the original sample's indexing;
  if $|R_k| < d$ **then**
    Pad R_k to size d with the smallest non-support indices in I_k according to a fixed ordering [14, Section IV-B];
  $I_{k+1} \leftarrow I_k \setminus R_k$;
Solve (6) on $\boldsymbol{z}_{I_\ell}$ to obtain $q_{\boldsymbol{z}_{I_\ell}}$;

In total, running Algorithm 2 requires ℓ solves of (6) and removes $(\ell - 1)d$ observations. The final predictor is

$$\hat{\Gamma}_\ell(\boldsymbol{z}) = \big\{z \in \mathcal{Z} : g\big(q_{\boldsymbol{z}_{I_\ell}}; z\big) \le 0\big\}.$$

Theorem 8 ([14, Theorem 4])**.** *Consider program (6) under Assumption 1. Fix $\ell \in \mathbb{N}_*$. Suppose $N \ge \ell d$. Let $\hat{\Gamma}_\ell$ be the predictor obtained by the support-removal cascade above. Assume that almost surely on $\boldsymbol{z} \sim \mathbb{P}^N$, every subsample in the cascade, including the final one, is nondegenerate. Then, for every $\varepsilon \in (0, 1)$,*

$$\mathbb{P}^N\{\hat{C}_\ell(\boldsymbol{z}) < 1 - \varepsilon\} \le B_{N,\ell d}(\varepsilon), \tag{21}$$

where $\hat{C}_\ell(\boldsymbol{z}) = \mathbb{P}[\hat{\Gamma}_\ell(\boldsymbol{z})]$.

Equality in (21) requires additional assumptions [14, Theorem 5]. The final optimizer need not violate every removed constraint; the certificate is specific to this cascade.

Appendix IX
Data generation

We adapt the vehicle-trajectory benchmark used by Lützow et al. [8], who in turn adopt the driving policies and point predictor of Tumu et al. [20, Section 4.2]. The implementation simulates the discrete-time kinematic bicycle model

$$x_{k+1} = x_k + \Delta v_k \cos\theta_k, \quad y_{k+1} = y_k + \Delta v_k \sin\theta_k,$$
$$\theta_{k+1} = \theta_k + \frac{\Delta v_k}{L}\tan\delta_k, \quad v_{k+1} = v_k + \Delta a_k,$$

where (x_k, y_k) is the position, θ_k is the heading, v_k is the speed, and (δ_k, a_k) are the steering and acceleration inputs. We set $\Delta = 0.1$, $L = 1$, and $(x_0, y_0, \theta_0, v_0) = (0, 0, 0, 0.5)$, and simulate 55 steps per trajectory.

For each trajectory, a left-turn, right-turn, or straight-driving policy is selected independently with probability $1/3$ each. All policies use nominal inputs $(\bar{\delta}_k, \bar{a}_k) = (0, 0.15)$ for $k = 0, \dots, 4$. For $k = 5, \dots, 54$, the nominal inputs are

respectively $(0.3, 0.18)$, $(-0.3, 0.18)$, and $(0, 0)$. The actual inputs satisfy

$$(\delta_k, a_k) = (\bar{\delta}_k, \bar{a}_k) + \eta_k, \qquad \eta_k \sim \mathcal{N}\big(0, \mathrm{diag}(0.3^2, 0.06^2)\big),$$

with independent noise across time steps and trajectories, independent of the policy choices.

The first five simulated states, indexed $1, \ldots, 5$, form the observed prefix. Starting from the last prefix state, a predictor extrapolates at constant speed v_5 with constant heading increment $\omega = (\theta_5 - \theta_1)/4$ per step:

$$\begin{aligned} \hat{p}_0 &= (x_5, y_5), \qquad \hat{\theta}_0 = \theta_5, \\ \hat{p}_{t+1} &= \hat{p}_t + \Delta v_5 (\cos\hat{\theta}_t, \sin\hat{\theta}_t), \\ \hat{\theta}_{t+1} &= \hat{\theta}_t + \omega. \end{aligned}$$

For a chosen horizon $T \in \{2, 30, 50\}$, the observation supplied to the conformal methods is the position residual

$$z = (x_{5+T}, y_{5+T}) - \hat{p}_T \in \mathbb{R}^2.$$

Calibration and test observations are generated from independent trajectories using this procedure.

APPENDIX X
PREDICTION SETS AND EMPIRICAL DISTRIBUTIONS

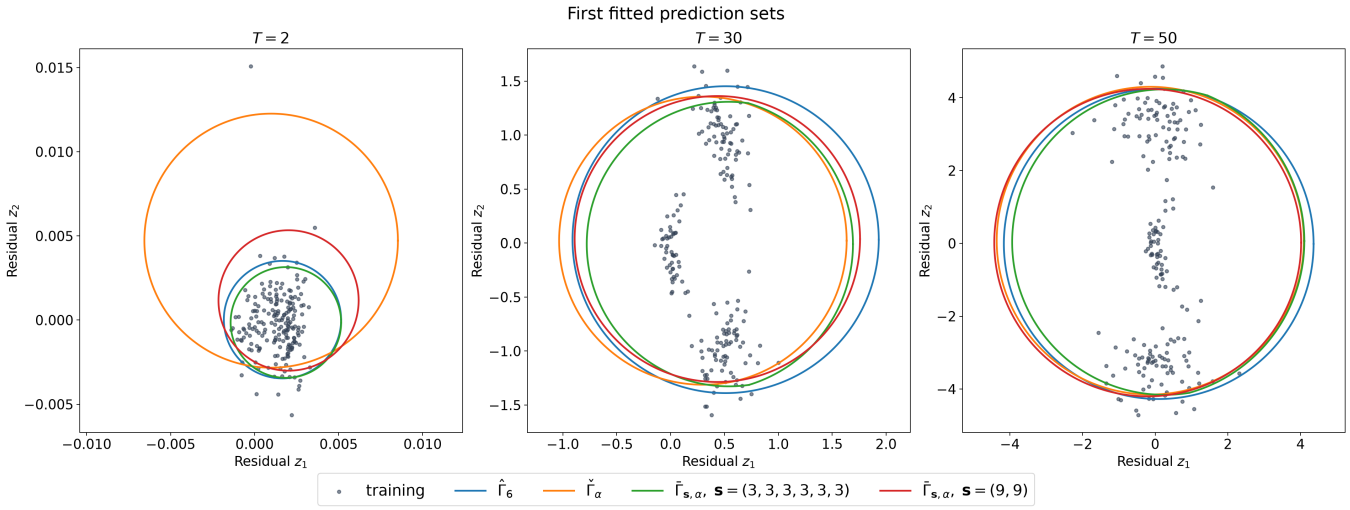


Fig. 1. Prediction sets for the first run, overlaid on the training residuals, at horizons $T = 2, 30, 50$ (left to right). Each panel has its own spatial scale. The intersection predictors retain the boundary arcs of their component disks.

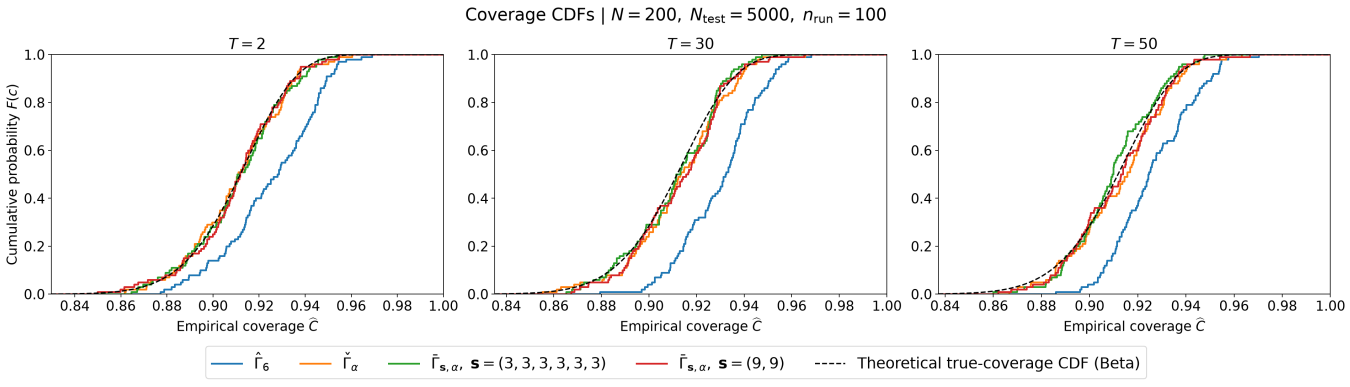


Fig. 2. Empirical cumulative distributions of estimated coverage across 100 runs. The dashed curve is the $\mathrm{Beta}(183, 18)$ true-coverage reference, exact for the proposed predictors under the stated assumptions and an upper bound on the true-coverage CDF for cascade removal. Finite test-sample estimation introduces additional noise.

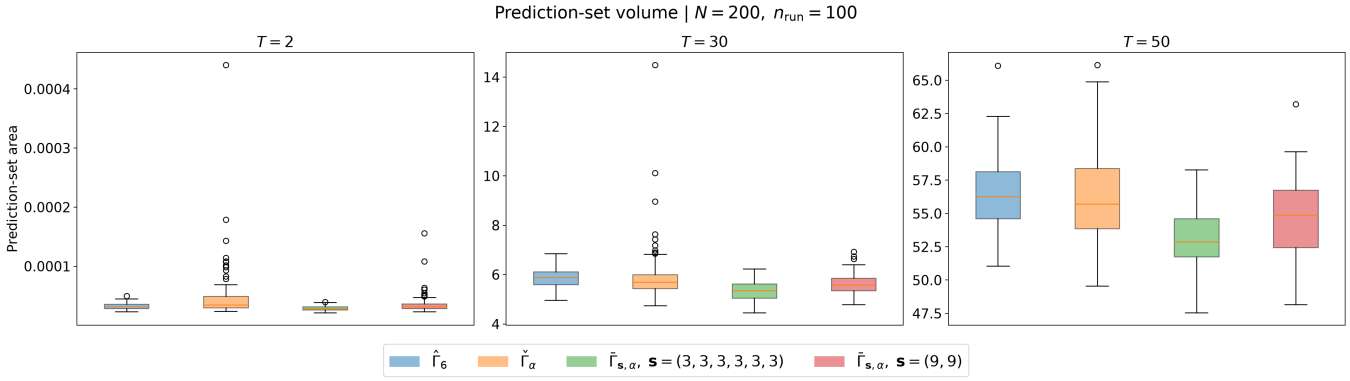


Fig. 3. Prediction-set areas across 100 runs at each horizon. Boxes show the interquartile range and median; whiskers extend to the most extreme observations within 1.5 interquartile ranges, with remaining observations plotted individually. Vertical scales differ between panels.

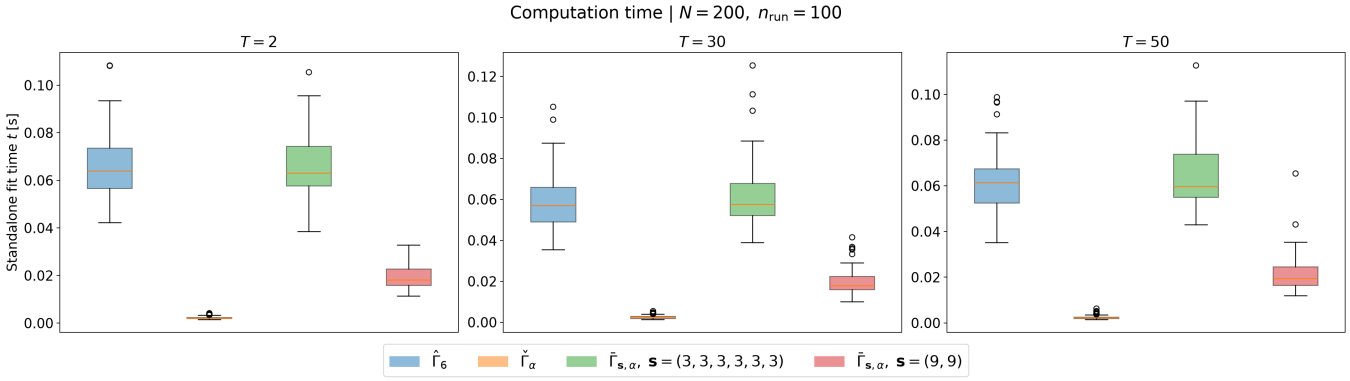


Fig. 4. Standalone fitting times across 100 runs, in seconds, using the same box-plot convention as Fig. 3. Times exclude test-set evaluation and area computation and describe the measured implementation.

APPENDIX XI
GEOMETRIC ILLUSTRATION OF THE CONSTRUCTION

Figure 5 illustrates the construction for smallest-enclosing-disk scenario programs, whose decision dimension is $d = 3$. At each stage, the optimizer fixes a disk; residual thresholding then shrinks this disk without another optimization solve. The selected batch contains the support observations and the observations with the largest remaining residuals. Removing this batch permits a new fit at the next stage, and the final predictor intersects the thresholded disks from all stages.

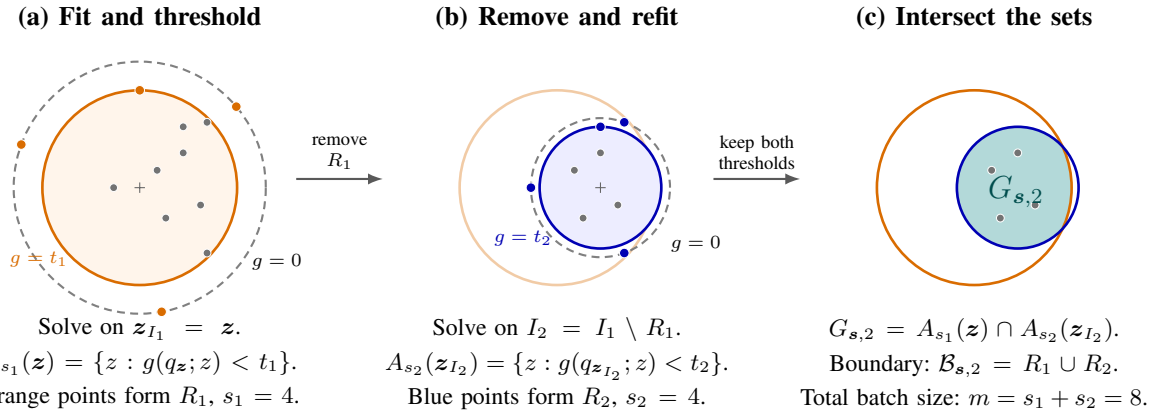


Fig. 5. Residual thresholding and cascade removal, illustrated with two stages and $s_1 = s_2 = 4$. Dashed circles are the fitted scenario boundaries $g = 0$; solid circles are the residual thresholds $g = t_k$. Each selected batch consists of three support observations and one additional observation defining the threshold. Crosses mark fitted centers, and gray points are retained observations. The shaded intersection is the final prediction set; its boundary includes arcs from both thresholded disks. Circle outlines depict the closures of the strict sublevel sets. Under the assumptions of Theorem 4, with $m = \lfloor \alpha(N+1) \rfloor$, this intersection agrees with the conformal predictor up to a $\mathbb{P}$-null set and has the exact coverage law $\mathbb{P}[G_{s,2}(z)] \sim \mathrm{Beta}(N-m+1, m) = \mathrm{Beta}(N-7, 8)$ for $z \sim \mathbb{P}^N$, since $m = s_1 + s_2 = 8$.